

CO323 Lab

E/21/019

Task 1

Task 1.1

1. Physical Connectivity and Cable Types

Objective: To establish a physical link between PC0 and PC1 without external network access.

- **Observation (RJ45 Direct/Straight-through Cable):** When connecting PC0 and PC1 directly using a standard RJ45 direct cable, the link fails to establish, and pings result in "Request Timed Out" or "Destination Host Unreachable".
- **Observation (Crossover Cable):** After switching to a crossover cable, connectivity is established, and the devices can successfully ping each other.
- **Reasoning:** PCs are similar devices (DTE to DTE). A straight-through cable does not cross the transmit and receive pairs, causing a signal mismatch where both devices try to "talk" on the same wire. A crossover cable correctly maps the transmit pins of one PC to the receive pins of the other.

2. Troubleshooting the Firewall Barrier

Observation: Even with the correct cable and matching subnets (192.168.100.1/24 and 192.168.100.2/24), the initial ping failed with "Request Timed Out".

- **Discovery:** The OS Firewall (such as Windows Defender) blocks incoming ICMP (Internet Control Message Protocol) packets by default for security.
- **Solution:** Disabling the firewall on both devices allows the ping to succeed.
- **Technical Logic:** A "Request Timed Out" indicates that the packet likely reached the destination, but the destination's security settings prevented it from processing the request or sending an Echo Reply back to the source.

```
C:\Users\hirus>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time=3ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Users\hirus>
```

No.	Time	Source	Destination	Protocol	Length	Info
37	09:00:26.902922	fe80::ad54:43aa:294...	ff02::fb	NDIS	102	Standard query 0x0000 PTR _googlecast._tcp.local, "Q"
38	09:00:27.000126	192.168.100.1	192.168.100.255	BROWSER	215	Request Announcement HS
39	09:00:27.762959	192.168.100.1	192.168.100.2	ICMP	74	Echo (ping) request id=0x0001, seq=48/12288, ttl=128 (reply in 40)
40	09:00:27.763782	192.168.100.2	192.168.100.1	ICMP	74	Echo (ping) reply id=0x0001, seq=48/12288, ttl=128 (request in 39)
41	09:00:28.518706	192.168.100.1	192.168.100.255	BROWSER	215	Request Announcement HS
42	09:00:28.781928	192.168.100.1	192.168.100.2	ICMP	74	Echo (ping) request id=0x0001, seq=49/12544, ttl=128 (reply in 43)
43	09:00:28.782847	192.168.100.2	192.168.100.1	ICMP	74	Echo (ping) reply id=0x0001, seq=49/12544, ttl=128 (request in 42)
44	09:00:29.188099	192.168.100.1	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
45	09:00:29.799485	192.168.100.1	192.168.100.2	ICMP	74	Echo (ping) request id=0x0001, seq=50/12800, ttl=128 (reply in 46)
46	09:00:29.800492	192.168.100.2	192.168.100.1	ICMP	74	Echo (ping) reply id=0x0001, seq=50/12800, ttl=128 (request in 45)
47	09:00:30.034392	192.168.100.1	192.168.100.255	BROWSER	215	Request Announcement HS
48	09:00:30.442417	192.168.100.1	224.0.0.251	NDIS	436	Standard query response 0x0000 TXT, cache flush PTR _ni-logos._tcp.local PTR HS._ni-logos._tcp.local SRV, cache flush 0 0 2343 HS...
49	09:00:30.814209	192.168.100.1	192.168.100.2	ICMP	74	Echo (ping) request id=0x0001, seq=51/13056, ttl=128 (reply in 50)
50	09:00:30.816043	192.168.100.2	192.168.100.1	ICMP	74	Echo (ping) reply id=0x0001, seq=51/13056, ttl=128 (request in 49)
51	09:00:31.546110	192.168.100.1	192.168.100.255	BROWSER	227	Browser Election Request
52	09:00:32.203937	192.168.100.1	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1

3. IP Address and Subnet Mask Combinations

Based on the lab instructions, the following tests were conducted on **PC0** while **PC1** remained constant at **192.168.100.2** with a **/24** subnet mask.

- **Scenario 1: PC0 IP 192.168.200.1, Subnet Mask /24**
 - **Observation:** Ping fails.
 - **Reasoning:** PC0 (192.168.200.0 network) and PC1 (192.168.100.0 network) are in different logical subnets. Without a router, traffic cannot pass between them as the devices do not consider each other "local."

```
C:\Users\hirus>ping 192.168.100.2

Pinging 192.168.100.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

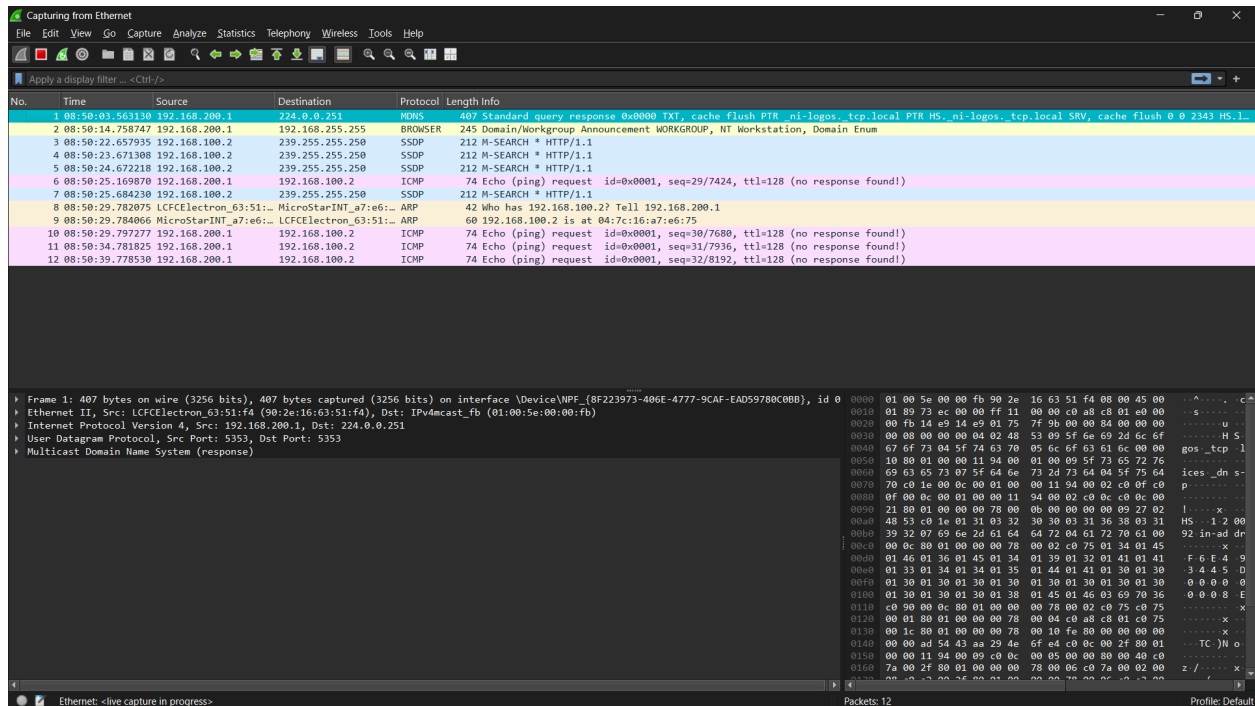
Ping statistics for 192.168.100.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- **Scenario 2: PC0 IP 192.168.200.1, Subnet Mask /16**
 - **Observation:** Ping fails.
 - **Reasoning:** This is a **subnet mismatch**. While PC0 (with /16) sees PC1 as local, PC1 (with /24) sees 192.168.200.1 as a remote network. When PC1 tries to reply, it looks for a "Default Gateway" that does not exist, causing the return packet to fail

```
C:\Users\hirus>ping 192.168.100.2

Pinging 192.168.100.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.100.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```



- **Scenario 3: PC0 IP 192.168.100.10, Subnet Mask /28**
 - **Observation:** Ping succeeds.
 - **Reasoning:** A /28 mask (255.255.255.240) for this IP creates a host range from 192.168.100.1 to 192.168.100.14. Since PC1 (192.168.100.2) falls within this range, they can communicate.

```
C:\Users\hirus>ping 192.168.100.2
```

```
Pinging 192.168.100.2 with 32 bytes of data:
```

```
Reply from 192.168.100.2: bytes=32 time=2ms TTL=128
```

```
Reply from 192.168.100.2: bytes=32 time=2ms TTL=128
```

```
Reply from 192.168.100.2: bytes=32 time=2ms TTL=128
```

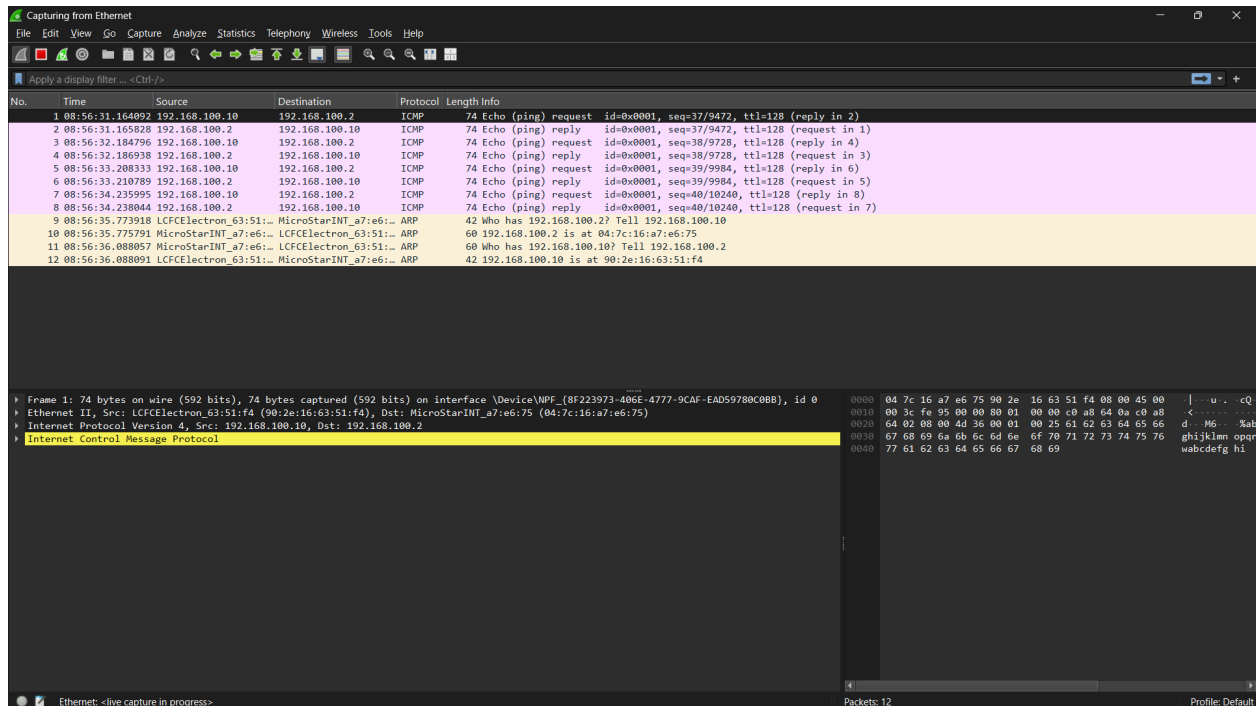
```
Reply from 192.168.100.2: bytes=32 time=2ms TTL=128
```

```
Ping statistics for 192.168.100.2:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 2ms, Maximum = 2ms, Average = 2ms
```



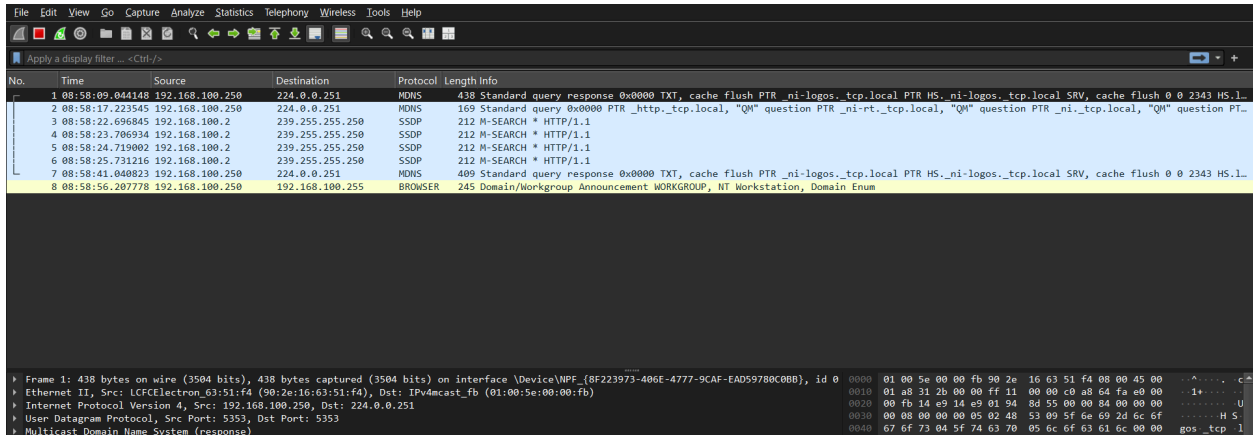
- **Scenario 4: PC0 IP 192.168.100.250, Subnet Mask /28**
 - **Observation:** Ping fails.
 - **Reasoning:** For this IP, a /28 mask defines a host range from 192.168.100.241 to 192.168.100.254. PC1 (192.168.100.2) is outside this boundary, causing the PC to assume the destination is on a different network.

```
C:\Users\hirus>ping 192.168.100.2

Pinging 192.168.100.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.100.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\hirus>
```



4. General Observations

- **Wireshark Monitoring:** All communications were observed through a packet-capturing device to distinguish between broadcast traffic (ARP) and direct unicast traffic (ICMP).
- **Promiscuous Mode:** Switching between modes was necessary to see all traffic on the wire versus only traffic destined for the specific host.
- **Cache Management:** Clearing the ARP cache was essential when changing IP addresses. Without clearing the cache, a PC might attempt to send data to a hardware MAC address that is no longer associated with that specific IP.

Task 1.2

When PCs are connected through a **switch**, IP addressing and subnet calculations remain unchanged, but **physical and data link layer behavior differ** from a direct cable connection.

```
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hirus>ping 192.168.100.2

Pinging 192.168.100.2 with 32 bytes of data:
Reply from 192.168.100.2: bytes=32 time=1ms TTL=128
Reply from 192.168.100.2: bytes=32 time=1ms TTL=128
Reply from 192.168.100.2: bytes=32 time<1ms TTL=128
Reply from 192.168.100.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.100.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\hirus>
```

Physical Layer

- A switch is an **intermediate Layer-2 device**.
- It internally handles signal crossing.
- Therefore, **standard straight-through Ethernet cables** can be used between PCs and the switch.

Data Link Layer (MAC Learning)

- The switch maintains a **MAC address table**.
- It learns the port of each PC by reading the **source MAC address** of incoming frames.
- **ARP requests** are broadcast to all ports.
- **ICMP unicast traffic** is forwarded only to the port where the destination PC is connected.

Wireshark Observations

- ARP broadcasts are visible on all PCs.
- ICMP unicast packets are visible **only on the sender and receiver**.
- Even in promiscuous mode, traffic between other PCs cannot be captured because the switch does not forward it to unrelated ports.

Cache Effects

- Restarting the switch clears its MAC table.
- After reset, the switch temporarily floods traffic until MAC addresses are relearned.
- Communication behavior depends on both the **PC ARP cache** and the **switch MAC table**.

TASK 2

Task 2.1

Configuration

- PC A: 192.168.1.1 /24
- PC B: 192.168.1.2 /24
- PC C: 192.168.1.3/24
- PC D: 192.168.1.4 /24

All PCs belong to the same broadcast domain.

(a) Ping Between All Devices

Observation

- All PCs can successfully ping each other.
- Initial pings show a slight delay.
- Wireshark captures show ARP Request (broadcast) followed by ARP Reply (unicast).

Explanation

- Since all devices are in the same subnet, no routing is required.
- The first ping triggers ARP to resolve IP → MAC addresses.
- After ARP resolution, ICMP Echo and Echo Reply packets are exchanged directly using unicast frames.

```

Ping statistics for 192.168.100.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\hirus> ping 192.168.100.2

```

```

Pinging 192.168.100.2 with 32 bytes of data:
Reply from 192.168.100.2: bytes=32 time=1ms TTL=128
Reply from 192.168.100.2: bytes=32 time=1ms TTL=128
Reply from 192.168.100.2: bytes=32 time=1ms TTL=128
Reply from 192.168.100.2: bytes=32 time=1ms TTL=128

```

```

Ping statistics for 192.168.100.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
PS C:\Users\hirus> ping 192.168.100.3

```

```

Pinging 192.168.100.3 with 32 bytes of data:
Reply from 192.168.100.3: bytes=32 time=1ms TTL=128
Reply from 192.168.100.3: bytes=32 time=2ms TTL=128
Reply from 192.168.100.3: bytes=32 time=1ms TTL=128
Reply from 192.168.100.3: bytes=32 time=1ms TTL=128

```

```

Ping statistics for 192.168.100.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
PS C:\Users\hirus> ping 192.168.100.4

```

```

Pinging 192.168.100.4 with 32 bytes of data:
Reply from 192.168.100.4: bytes=32 time=1ms TTL=128
Reply from 192.168.100.4: bytes=32 time=1ms TTL=128
Reply from 192.168.100.4: bytes=32 time=1ms TTL=128
Reply from 192.168.100.4: bytes=32 time=1ms TTL=128

```

```

Ping statistics for 192.168.100.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
PS C:\Users\hirus> |

```

No.	Time	Source	Destination	Protocol	Length	Info
7	08:37:53.246400	192.168.100.2	192.168.100.1	ICMP	74	Echo (ping) reply id=0x0001, seq=63/16128, ttl=128 (request in 6)
8	08:37:53.518538	Cisco 7f:26:13	Spanning-tree (for-...	STP	60	Conf. Root = 32768/1/00:13:80:7f:26:00 Cost = 0 Port = 0x8013
9	08:37:54.259046	192.168.100.1	192.168.100.2	ICMP	74	Echo (ping) request id=0x0001, seq=64/16384, ttl=128 (reply in 10)
10	08:37:54.260478	192.168.100.2	192.168.100.1	ICMP	74	Echo (ping) reply id=0x0001, seq=64/16384, ttl=128 (request in 9)
11	08:37:55.265540	192.168.100.1	192.168.100.2	ICMP	74	Echo (ping) request id=0x0001, seq=65/16640, ttl=128 (reply in 12)
12	08:37:55.266991	192.168.100.2	192.168.100.1	ICMP	74	Echo (ping) reply id=0x0001, seq=65/16640, ttl=128 (request in 11)
13	08:37:55.518539	Cisco 7f:26:13	Spanning-tree (for-...	STP	60	Conf. Root = 32768/1/00:13:80:7f:26:00 Cost = 0 Port = 0x8013
14	08:37:55.769040	Cisco 7f:26:13	Cisco 7f:26:13	LOOP	60	Reply
15	08:37:56.778699	LCFCElectron_63:51:...	MicroStarINT_a7:e6:...	ARP	42	Who has 192.168.100.2? Tell 192.168.100.1
16	08:37:56.780308	MicroStarINT_a7:e6:...	LCFCElectron_63:51:...	ARP	60	192.168.100.2 is at 04:7c:16:a7:e6:75
17	08:37:56.853658	MicroStarINT_a7:e6:...	LCFCElectron_63:51:...	ARP	60	Who has 192.168.100.1? Tell 192.168.100.2
18	08:37:56.853691	LCFCElectron_63:51:...	MicroStarINT_a7:e6:...	ARP	42	192.168.100.1 is at 90:2e:16:63:51:f4
19	08:37:56.914182	192.168.100.4	192.168.100.255	NBNS	92	Name query NB DESKTOP-HIM509A1c<>
20	08:37:57.517891	Cisco 7f:26:13	Spanning-tree (for-...	STP	60	Conf. Root = 32768/1/00:13:80:7f:26:00 Cost = 0 Port = 0x8013
21	08:37:57.677953	192.168.100.4	192.168.100.255	NBNS	92	Name query NB DESKTOP-HIM509A1c<>
22	08:37:58.442385	192.168.100.4	192.168.100.255	NBNS	92	Name query NB DESKTOP-HIM509A1c<>
23	08:37:58.589118	192.168.100.1	192.168.100.3	ICMP	74	Echo (ping) request id=0x0001, seq=66/16896, ttl=128 (reply in 24)
24	08:37:58.590143	192.168.100.3	192.168.100.1	ICMP	74	Echo (ping) reply id=0x0001, seq=66/16896, ttl=128 (request in 23)
25	08:37:59.517979	Cisco 7f:26:13	Spanning-tree (for-...	STP	60	Conf. Root = 32768/1/00:13:80:7f:26:00 Cost = 0 Port = 0x8013
26	08:37:59.599306	192.168.100.1	192.168.100.3	ICMP	74	Echo (ping) request id=0x0001, seq=67/17152, ttl=128 (reply in 27)
27	08:37:59.600115	192.168.100.3	192.168.100.1	ICMP	74	Echo (ping) reply id=0x0001, seq=67/17152, ttl=128 (request in 26)
28	08:38:00.544852	192.168.100.4	192.168.100.3	ICMP	74	Echo (ping) request id=0x0001, seq=68/17408, ttl=128 (reply in 29)

(b) Change IP Address of One PC (PC C)

Step 1: Ping from Other PCs to C

Observation

- Initial pings to PC C fail and display a destination unreachable error.
- Wireshark shows ARP requests with no valid reply (or stale ARP entries).

Reasoning

- Other PCs still have the old IP–MAC mapping for PC C in their ARP cache.
- Since PC C's IP has changed, ARP resolution fails until caches are updated.

Step 2: Ping from PC C to Others

Observation

- Pings from PC C to other PCs succeed.
- ARP requests are generated by PC C.

Reasoning

- PC C initiates fresh ARP requests using its new IP.
- Other PCs respond, updating their ARP tables accordingly.

Step 3: Ping Again from Others to C

Observation

- Pings now succeed.

Explanation

- ARP tables on all PCs are updated due to PC C's outgoing traffic.
- This demonstrates how bidirectional communication refreshes ARP caches.

(c) Change Switch Port of One PC (PC A)

Step 1: Ping from Others to A

Observation

- First ping may fail or experience delay.
- Wireshark shows ARP broadcast.

Reasoning

- The switch still associates PC A's MAC address with the old port.
- Until the switch relearns the MAC-to-port mapping, frames may be misdirected.

Step 2: Ping from A to Others

Observation

- Pings succeed.

Explanation

- When PC A sends frames, the switch learns its MAC address on the new port.
- The MAC address table is updated immediately.

Step 3: Ping Again from Others to A

Observation

- Pings succeed without delay.

Explanation

- The switch has updated its forwarding table.
- Communication resumes normally using unicast frames.

Task 2.2

Configuration

- **Subnet A**
 - PC A: 192.168.1.2 /24
 - PC B: 192.168.1.3 /24
- **Subnet B**
 - PC C: 192.168.2.2 /24
 - PC D: 192.168.2.3 /24
- **Subnet C**
 - PC E: 192.168.3.1 /24

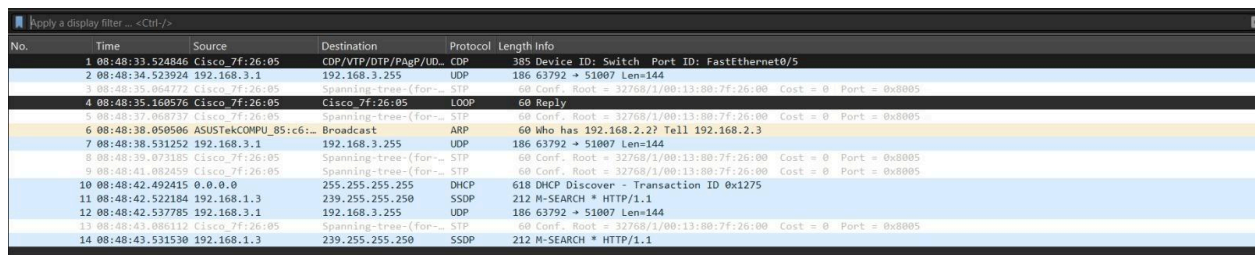
(a) Ping Within Each Subnet

Observation

- PCs within the same subnet can ping each other successfully.
- Wireshark shows ARP and ICMP packets.

Promiscuous vs Non-Promiscuous Mode

- Promiscuous mode shows broadcast ARP packets from all PCs.



No.	Time	Source	Destination	Protocol	Length	Info
1	08:48:33.524846	Cisco-7f:26:05	CDP/VTP/DTP/PAgP/UD...	CDP	385	Device ID: Switch Port ID: FastEthernet0/5
2	08:48:34.523924	192.168.3.1	192.168.3.255	UDP	186	63792 → 51007 Len=144
3	08:48:35.064752	Cisco-7f:26:05	Spanning-tree (for...	STP	60	Conf. Root = 32768/1/00:13:00:7f:26:00 Cost = 0 Port = 0x8005
4	08:48:35.160576	Cisco-7f:26:05	Cisco-7f:26:05	LOOP	60	Reply
5	08:48:37.000757	Cisco-7f:26:05	Spanning-tree (for...	STP	60	Conf. Root = 32768/1/00:13:00:7f:26:00 Cost = 0 Port = 0x8005
6	08:48:38.050506	ASUSTekCOMPU_85:c6:...	Broadcast	ARP	60	Who has 192.168.2.2? Tell 192.168.2.3
7	08:48:38.531252	192.168.3.1	192.168.3.255	UDP	186	63792 → 51007 Len=144
8	08:48:39.073185	Cisco-7f:26:05	Spanning-tree (for...	STP	60	Conf. Root = 32768/1/00:13:00:7f:26:00 Cost = 0 Port = 0x8005
9	08:48:41.002459	Cisco-7f:26:05	Spanning-tree (for...	STP	60	Conf. Root = 32768/1/00:13:00:7f:26:00 Cost = 0 Port = 0x8005
10	08:48:42.492415	0.0.0.0	255.255.255.255	DHCP	618	DHCP Discover - Transaction ID 0x1275
11	08:48:42.521884	192.168.1.3	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1
12	08:48:42.537785	192.168.3.1	192.168.3.255	UDP	186	63792 → 51007 Len=144
13	08:48:43.086112	Cisco-7f:26:05	Spanning-tree (for...	STP	60	Conf. Root = 32768/1/00:13:00:7f:26:00 Cost = 0 Port = 0x8005
14	08:48:43.531530	192.168.1.3	239.255.255.250	SSDP	212	M-SEARCH * HTTP/1.1

- Non-promiscuous mode only shows frames destined for the capturing PC.

Explanation

- Devices in the same subnet rely on ARP and Layer-2 switching.
- The switch floods broadcasts but forwards unicasts selectively.

```
C:\Users\hirus>ping 192.168.1.3
```

```
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=2ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
```

```
Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

```
C:\Users\hirus>
```

577	08:43:26.500167	Cisco	7f:26:05	Spanning-tree-(for-...	STP	60	Conf. TC + Root = 32768/1/00:13:80:7f:26:00 Cost = 0 Port = 0x8005
578	08:43:26.580249	Cisco	7f:26:05	CDP/VTP/DTP/PagP/UD...	CDP	385	Device ID: Switch Port ID: FastEthernet0/5
579	08:43:26.627368	fe80::ad54:43aa:294...	ff02::1:2	DHCPv6	120	Information-request XID: 0xc0a207 CID: 000100012a57d9a6902e166351f4	
580	08:43:27.063959	192.168.1.3	192.168.1.2	ICMP	74	Echo (ping) request id=0x0001, seq=8/2048, ttl=128 (reply in 581)	
581	08:43:27.063241	192.168.1.2	192.168.1.3	ICMP	74	Echo (ping) reply id=0x0001, seq=8/2048, ttl=128 (request in 580)	
582	08:43:28.066412	192.168.1.3	192.168.1.2	ICMP	74	Echo (ping) request id=0x0001, seq=9/2304, ttl=128 (reply in 583)	
583	08:43:28.066561	192.168.1.2	192.168.1.3	ICMP	74	Echo (ping) reply id=0x0001, seq=9/2304, ttl=128 (request in 582)	
584	08:43:28.501130	Cisco	7f:26:05	Spanning-tree-(for-...	STP	60	Conf. TC + Root = 32768/1/00:13:80:7f:26:00 Cost = 0 Port = 0x8005
585	08:43:29.078409	192.168.1.3	192.168.1.2	ICMP	74	Echo (ping) request id=0x0001, seq=10/2560, ttl=128 (reply in 586)	
586	08:43:29.078571	192.168.1.2	192.168.1.3	ICMP	74	Echo (ping) reply id=0x0001, seq=10/2560, ttl=128 (request in 585)	
587	08:43:29.946577	192.168.3.1	192.168.3.255	UDP	186	63792 + 51007 Len=144	
588	08:43:30.090455	192.168.1.3	192.168.1.2	ICMP	74	Echo (ping) request id=0x0001, seq=11/2816, ttl=128 (reply in 589)	
589	08:43:30.090620	192.168.1.2	192.168.1.3	ICMP	74	Echo (ping) reply id=0x0001, seq=11/2816, ttl=128 (request in 588)	
590	08:43:30.506986	Cisco	7f:26:05	Spanning-tree-(for-...	STP	60	Conf. TC + Root = 32768/1/00:13:80:7f:26:00 Cost = 0 Port = 0x8005
591	08:43:31.510990	Cisco	7f:26:05	Spanning-tree-(for-...	STP	60	Conf. TC + Root = 32768/1/00:13:80:7f:26:00 Cost = 0 Port = 0x8005
592	08:43:33.362663	192.168.1.2	224.0.0.251	MDNS	405	Standard query response 0x0000 TXT, cache flush PTR _ni-logs._tcp.local PTR HS._ni-logs._tcp.local SRV, cache flush 0 0 2343 H	
593	08:43:33.519077	Cisco	7f:26:05	CDP/VTP/DTP/PagP/UD...	CDP	385	Device ID: Switch Port ID: FastEthernet0/5
594	08:43:33.956012	192.168.3.1	192.168.3.255	UDP	186	63792 + 51007 Len=144	
523	08:43:07.263012	192.168.3.1	224.0.0.251	MDNS	112	Standard query 0x0000 PTR _sleep-proxy._udp.local, "QM" question OPT	
524	08:43:07.636716	192.168.3.1	224.0.0.251	MDNS	460	Standard query response 0x0000 TXT, cache flush PTR _ni-logs._tcp.local PTR DESKTOP-0LC733D._ni-logs._tcp.local SRV, cache	
525	08:43:07.810506	LCFCElectron_63:51::	Broadcast	ARP	42	Who has 192.168.1.3? Tell 192.168.1.2	
526	08:43:07.810969	MicroStarINT_a7:e6::	ARP	60	192.168.1.3 is at 047c:16:a7:e6:75		
527	08:43:07.810982	192.168.1.2	192.168.1.3	ICMP	74	Echo (ping) request id=0x0001, seq=17/4352, ttl=128 (reply in 528)	
528	08:43:07.813313	192.168.1.3	192.168.1.2	ICMP	74	Echo (ping) reply id=0x0001, seq=17/4352, ttl=128 (request in 527)	
529	08:43:07.825599	192.168.3.1	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question	
530	08:43:07.825617	fe80::d7a2:455f:dfa...	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question	
531	08:43:07.825850	fe80::d7a2:455f:dfa...	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question	
532	08:43:07.826069	192.168.3.1	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question	
533	08:43:08.462287	Cisco	7f:26:05	Spanning-tree-(for-...	STP	60	Conf. TC + Root = 32768/1/00:13:80:7f:26:00 Cost = 0 Port = 0x8005
534	08:43:08.794328	192.168.3.1	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1	
535	08:43:08.828879	192.168.1.2	192.168.1.3	ICMP	74	Echo (ping) request id=0x0001, seq=18/4608, ttl=128 (reply in 536)	
536	08:43:08.830239	192.168.1.3	192.168.1.2	ICMP	74	Echo (ping) reply id=0x0001, seq=18/4608, ttl=128 (request in 535)	
537	08:43:09.854023	192.168.1.2	192.168.1.3	ICMP	74	Echo (ping) request id=0x0001, seq=19/4864, ttl=128 (reply in 538)	
538	08:43:09.854873	192.168.1.3	192.168.1.2	ICMP	74	Echo (ping) reply id=0x0001, seq=19/4864, ttl=128 (request in 537)	
539	08:43:09.904006	192.168.3.1	192.168.3.255	UDP	186	63792 + 51007 Len=144	
540	08:43:10.467094	Cisco	7f:26:05	Spanning-tree-(for-...	STP	60	Conf. TC + Root = 32768/1/00:13:80:7f:26:00 Cost = 0 Port = 0x8005
541	08:43:10.868876	192.168.3.1	224.0.0.251	MDNS	511	Standard query response 0x0000 TXT, cache flush PTR _nvstream_dbd._tcp.local PTR 3.21.0.36-DESKTOP-0LC733D.620d3a98-3d75-480	
542	08:43:10.877221	192.168.1.2	192.168.1.3	ICMP	74	Echo (ping) request id=0x0001, seq=20/5120, ttl=128 (reply in 543)	
543	08:43:10.878604	192.168.1.3	192.168.1.2	ICMP	74	Echo (ping) reply id=0x0001, seq=20/5120, ttl=128 (request in 542)	
544	08:43:11.803537	192.168.3.1	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1	
545	08:43:11.818485	192.168.3.1	224.0.0.251	MDNS	460	Standard query response 0x0000 TXT, cache flush PTR _ni-logs._tcp.local PTR DESKTOP-0LC733D._ni-logs._tcp.local SRV, cache	
546	08:43:12.470642	Cisco	7f:26:05	Spanning-tree-(for-...	STP	60	Conf. TC + Root = 32768/1/00:13:80:7f:26:00 Cost = 0 Port = 0x8005
547	08:43:12.480041	MicroStarINT_a7:e6::	LCFCElectron_63:51::	ARP	60	Who has 192.168.1.2? Tell 192.168.1.3	
548	08:43:12.480042	MicroStarINT_a7:e6::	LCFCElectron_63:51::	ARP	42	Who has 192.168.1.2? Tell 192.168.1.3	

Ping Between Different Subnets

Observation

- Ping fails between PCs in different subnets.
- Wireshark shows repeated ARP requests or ICMP destination unreachable messages.

Reasoning

- Hosts attempt to send packets to a default gateway.
- Since no router is present, inter-subnet communication is impossible.
- A Layer-2 switch cannot perform routing.

```
Pinging 192.168.2.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\hirus>
```

(b) Change the Subnet of One PC

Observation

- PC E can successfully ping other PCs in Subnet B.

```
C:\Users\Arnikan>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Arnikan>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time=1ms TTL=128
Reply from 192.168.2.2: bytes=32 time=1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

- Other PCs in Subnet B cannot ping PC E initially.

```
Pinging 192.168.2.4 with 32 bytes of data:  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.  
  
Ping statistics for 192.168.2.4:  
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Explanation (ARP Cache Effect)

- After the subnet change, PC E initiates communication using its **new IP address**, sending fresh ARP requests.
- PCs in Subnet B still hold **stale ARP cache entries** that associate the old IP–MAC mapping (or another device) with that address.
- Because of this outdated ARP information, ICMP Echo Requests sent to PC E are delivered to an incorrect or non-existent MAC address, causing ping failures.

Why PC E → Others Works First

- When PC E sends packets, the switch learns PC E's MAC address on the correct port.
- The destination PCs reply, but their ARP tables are only updated after they see PC E's ARP or traffic.

Resolution

- Once ARP caches on Subnet B PCs are refreshed (either automatically, by clearing ARP cache, or after sufficient bidirectional traffic), pings to PC E succeed.